

Contents

1	Brownian Motion in Soft Confinement	3
1.1	Introduction	3
1.1.1	Deterministic soft lubrication	3
1.1.2	Stochastic soft lubrication	3
1.2	Toy Model for soft Brownian Motion	4
1.2.1	Overview of the Toy Model	4
1.3	Theoretical developments	5
1.4	Numerical modelling	8
1.4.1	Discretisation	8
1.4.2	Time step selection	8
1.5	Results	9
1.6	Conclusion and perspectives on the chapter	9

Chapter 1

Brownian Motion in Soft Confinement

1.1 Introduction

In many biological systems, one must not only take into account Brownian behaviours, but also carefully evaluate the effect of the system's or parts of the system's deformability on its dynamics. Indeed, if we take the example of a lipid bilayer, which is a soft and quasi-two-dimensional fluid, the thermal energy $k_B\Theta$ of the system is sufficient to cause long-wavelength bending fluctuations of the cell membrane, thickness fluctuations, lipid movement within the plane itself, and so forth. Naturally, a neighbouring fluid's motion is then affected by these shape modifications, which can in turn affect the dynamics of a body evolving in that same bath. This coupling is not trivial. I describe in this chapter what has already been developed on that matter, starting with the more well-established case of deterministic soft lubrication. I then present an overview of the stochastic case, where bodies or confinements are fluctuating and deformable. The remainder of the chapter is articulated around the presentation, development and numerical verifications of a simple and elementary model we developed during my doctoral contract to assess this same problematique of soft Brownian effects.

1.1.1 Deterministic soft lubrication

1.1.2 Stochastic soft lubrication

The aim of this chapter is to probe at these modifications to colloid's behaviour.

Because of the level of complexity of such a study, I present in this chapter a most simple model, with elementary ingredients, which can still qualitatively capture all the effects we are interested in. Such model is detailed in the next section. In the following sections, I develop the numerical model used to simulate this toy model. I then present the theoretical aspects of the model, developed by David S. Dean. Finally, I show how numerics and theory compare.

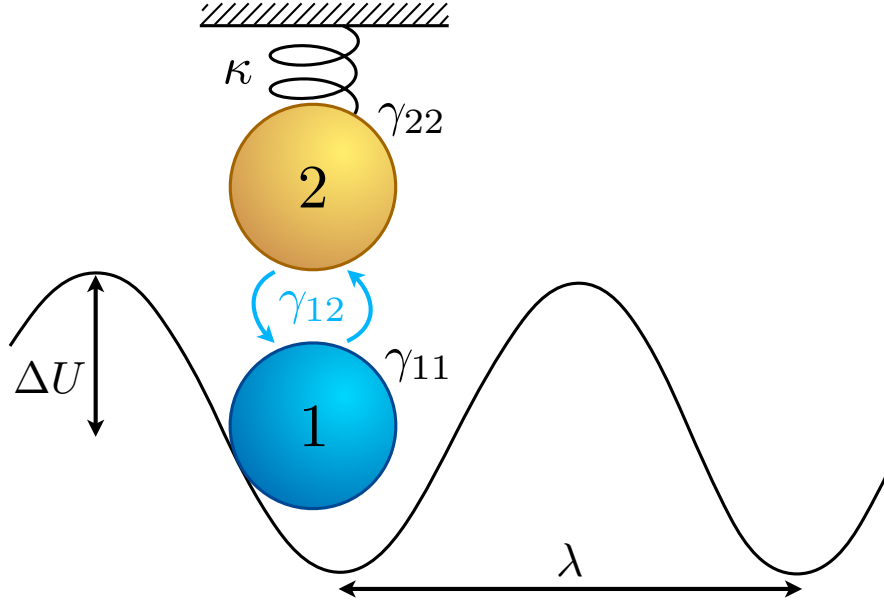


Figure 1.1: Representation of the system. A Brownian colloid of interest (blue) is trapped in a periodic potential of amplitude ΔU and spatial period λ . An elastic membrane is modelled as a second Brownian colloid (orange) subject to a harmonic potential of inverse stiffness of compliance κ . This membrane is coupled to the colloid of interest through an indirect hydrodynamic coupling represented by the coefficient γ_{12} , and both the colloid and the membrane feel their own self-friction γ_{11} and γ_{22} , respectively.

1.2 Toy Model for soft Brownian Motion

1.2.1 Overview of the Toy Model

From here on, we study a purely one-dimensional case of a Brownian colloid coupled to an elastic and fluctuating membrane (Fig. 1.1). The Brownian colloid is represented by a first degree of freedom, q_1 , and is submitted to a periodic trapping potential $\phi(q_1)$ of spatial period λ and amplitude ΔU . We couple to this colloid the toy model version of the membrane. We represent the membrane with a colloid that is submitted in a harmonic potential of energy $\frac{q_2^2}{2\kappa}$, where κ is the compliance (or inverse stiffness). It represents the elasticity of the membrane. This second colloid is tracked with the second degree of freedom, q_2 .

Both the Brownian colloid and the membrane feel their own self-friction, respectively γ_{11} and γ_{22} . Furthermore, they interact through an indirect, hydrodynamic-like interaction γ_{12} . This interaction represents the effect of the fluid that is displaced by the motion of the colloid on the membrane, and *vice-versa*. All together, these terms form the friction matrix $\mathbf{\Gamma}$

$$\mathbf{\Gamma} = \begin{pmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{12} & \gamma_{22} \end{pmatrix} \quad (1.1)$$

which can be inverted to give the mobility matrix of the system, as $\mathbf{M} = \mathbf{\Gamma}^{-1}$. Both matrices must be positive definite, and, because their diagonal terms account for the self-friction and self-mobility respectively, $\mu_{11} > 0$ and $\mu_{22} > 0$.

We can write the total energy as

$$H(q_1, q_2) = \phi(q_1) + \frac{q_2^2}{2\kappa}, \quad (1.2)$$

and we notice the absence of Hamiltonian interactions in the system. In that case, we work in the overdamped limit, and we write the coupled Langevin equations describing the evolutions of q_1 and q_2 as

$$\gamma_{11} \frac{dq_1}{dt} + \gamma_{12} \frac{dq_2}{dt} = -\phi'(q_1) + \eta_1(t), \quad (1.3)$$

$$\gamma_{12} \frac{dq_1}{dt} + \gamma_{22} \frac{dq_2}{dt} = -\frac{q_2}{\kappa} + \eta_2(t), \quad (1.4)$$

where η_i and η_j (with $j \in \{1, 2\}$) are Gaussian white noises of zero mean and correlation functions $\langle \eta_i(t) \eta_j(t') \rangle = 2k_B T \delta(t - t') \gamma_{ij}$.

$$P_{\text{eq}} = \frac{\exp[-\beta H(q_1, q_2)]}{Z}, \quad (1.5)$$

1.3 Theoretical developments

The objective of the following developments is to compute the modification to the long-time dynamics of the Brownian colloid diffusing within the sinusoidal trap. The development I choose to present here is not the most rigorous, and corresponds to the very stiff ($\kappa \rightarrow 0$) limit of the elastic membrane. The interested reader can find a more rigorous argument in the appendices of this thesis, alongside a second development for the very compliant limit ($\kappa \rightarrow \infty$).

We begin by refactoring Eqs. 1.3 and 1.4 in the deterministic case $\Theta = 0$

$$\frac{dq_1}{dt} = -\mu_{11} \phi'(q_1) - \frac{\mu_{12}}{\kappa} q_2, \quad (1.6)$$

$$\frac{dq_2}{dt} = -\mu_{12} \phi'(q_1) - \frac{\mu_{22}}{\kappa} q_2. \quad (1.7)$$

We may then integrate to express q_2 as

$$q_2 = -\frac{\kappa \mu_{12}}{\mu_{22}} \left(1 + \frac{\kappa}{\mu_{22}} \frac{d}{dt} \right)^{-1} \phi'(q_1). \quad (1.8)$$

In the small κ limit, we then expand the right-hand bracket term, taking into account the chain rule

$$q_2 \simeq -\frac{\kappa\mu_{12}}{\mu_{22}} \left[\phi'(q_1) - \frac{\kappa}{\mu_{22}} \phi''(q_1) \frac{dq_1}{dt} + O(\tau_2^2) \right]. \quad (1.9)$$

Here, we note that τ_2 is the characteristic relaxation time of the harmonic trapping corresponding to the membrane elasticity, as $\tau_2 = \frac{\kappa}{\mu_{22}}$. Now, substituting this new expression for q_2 in Eq. 1.6, we obtain

$$\frac{dq_1}{dt} = - \left(\mu_{11} - \frac{\mu_{12}^2}{\mu_{22}} \right) \phi'(q_1(t)) - \frac{\kappa\mu_{12}^2}{\mu_{22}^2} \phi''(q_1(t)) \frac{dq_1}{dt}, \quad (1.10)$$

which we can reexpress as

$$\frac{dq_1}{dt} = -\mu_e(q_1) \phi'(q_1), \quad (1.11)$$

where μ_e is a dressed mobility with expression

$$\mu_e = \frac{\mu_{11}\mu_{22} - \mu_{12}^2}{\mu_{22} \left(1 + \frac{\kappa\phi''(q_1)\mu_{12}^2}{\mu_{22}^2} \right)}. \quad (1.12)$$

Remembering that $\mathbf{\Gamma} = \mathbf{M}^{-1}$,

$$\mu_e = \frac{1}{\gamma_{11} \left(1 + \frac{\kappa\phi''(q_1)\mu_{12}^2}{\mu_{11}^2} \right)}. \quad (1.13)$$

Now, we can avoid reexpressing the noise should be added to Eq. 1.11 for the thermal case by working directly with Fokker-Planck formalism. This formalism is introduced in its elementary form at the beginning of my thesis, and developed for more complex cases in Chapter 2. **CHECK CHAPT NR AT THE END OF THE REDACTION HERE** Doing such a thing is not rigorous, as I previously mentioned, because the effective stochastic process has not been derived. Eliminating fast variables in Eq. 1.11 typically introduces non-trivial corrections. However, since we assume separation of timescales, the effective slow process still permits the writing of a uniquely-defined Fokker-Planck equation.

This Fokker-Planck equation reads

$$\frac{\partial P_1(q_1, t)}{\partial t} = \frac{\partial}{\partial q_1} \mu_e(q_1) \left[k_B \Theta \frac{\partial P_1(q_1, t)}{\partial q_1} + \phi'(q_1) P_1(q_1, t) \right] \quad (1.14)$$

and, noting that the time derivative of the probability density function $P_1(q_1, t)$ can be expressed in terms of probability flux $J(q_1, t)$,

$$\frac{\partial P_1(q_1, t)}{\partial t} = -\frac{\partial J}{\partial q_1}. \quad (1.15)$$

This allows us to write

$$J(q_1, t) = -D_e(q_1) \frac{\partial}{\partial q_1} P_1(q_1, t) - \mu_e \phi'(q_1) P_1(q_1, t), \quad (1.16)$$

where $D_e = k_B \Theta$. The above equation can be rewritten

$$J = -D_e(q_1) \exp[-\beta\phi(q_1)] \frac{d}{dq} \{ \exp[\beta\phi(q_1)] P_1(q_1, t) \} \quad (1.17)$$

with $\beta = \frac{1}{k_B \Theta}$.

Now, we tilt the potential by adding a small force F , which gives a new potential $U(q_1)$

$$U(q_1) = \phi(q_1) - Fq_1 . \quad (1.18)$$

Then, the current equation can be rewritten taking account this new force, and gives

$$\frac{d}{dq_1} \{ \exp[\beta U(q_1)] P(q_1, t) \} = -\frac{J(q_1, t)}{D_e(q_1)} \exp[\beta U(q_1)] . \quad (1.19)$$

We can then integrate over a period λ , and, using periodicity, we retrieve

$$P_1(q_1, t) = \frac{J(q_1) \exp[-\beta U(q_1)]}{1 - \exp(-\beta F \lambda)} \int_{q_1}^{q_1 + \lambda} \frac{\exp[\beta U(q)]}{D_e(q)} dq . \quad (1.20)$$

check $J(q, t)$ or just q .

By essence, $P_1(q_1, t)$ is normalised

$$\int_0^\lambda P_1(q_1) dq_1 = 1 , \quad (1.21)$$

and thus we retrieve for J

$$J(q_1) = \frac{1 - \exp[-\beta F \lambda]}{\int_0^\lambda \exp[-\beta U(q)] dq \int_q^{q+\lambda} \frac{\exp[\beta U(q)]}{D_e(q)} dq} . \quad (1.22)$$

In the small force limit $F \rightarrow 0$,

$$1 - \exp(-\beta F \lambda) \simeq \beta F \lambda , \quad (1.23)$$

which, to leading order in F , gives

$$J(q_1) = \frac{\beta F \lambda}{\int_0^\lambda \exp[-\beta\phi(q_1)] dq_1 \int_0^\lambda \frac{\exp[\beta\phi(q_1)]}{D_e(q_1)} dq_1} . \quad (1.24)$$

Finally, we note that the drift velocity should read $v = J(q_1)\lambda$, and that the effective diffusion coefficient over λ should be

$$D^* = \frac{k_B \Theta v}{F} = \frac{\lambda^2}{\int_0^\lambda \exp[-\beta\phi(q_1)] dq_1 \int_0^\lambda \frac{\exp(\beta\phi(q_1))}{D_e(q_1)} dq_1} , \quad (1.25)$$

which is the generalised Lifson-Jackson formula. cite.

Finally, we replace $D_e(q_1)$ in the expression

$$D^* = \frac{k_B \Theta \lambda^2}{\gamma_{11} \int_0^\lambda dq \exp[\beta \phi(q)] \left(1 - \frac{\kappa \gamma_{12}^2 \phi'^2(q)}{k_B \Theta \gamma_{11}^2}\right) \int_0^\lambda dq \exp[-\beta \phi(q)]} \quad (1.26)$$

And, by expansion in very small κ , and setting $\phi(q_1) = \Delta U \phi(\frac{q_1}{\lambda})$, where ΔU is the amplitude of the periodic trap

$$D^* = \frac{k_B \Theta}{\gamma_{11} \int_0^1 dp \exp[\beta \Delta U \psi(p)] \left(1 - \frac{\kappa \gamma_{12}^2 \Delta U^2 \psi'^2(p)}{k_B \Theta \gamma_{11}^2}\right) \int_0^1 dp \exp[-\beta \Delta U \psi(p)]} . \quad (1.27)$$

here.

1.4 Numerical modelling

1.4.1 Discretisation

My main mission in this project is to simulate the above Eqs. 1.3 and 1.4 efficiently. To do so, I discretise these equations using a simple forward Euler scheme (which was introduced in the beginning of this thesis). The discretised equations read

$$q_1(t + \Delta t) - q_1(t) = -\mu_{11} \phi'(q_1(t)) \Delta t - \frac{\mu_{12}}{\kappa} q_2(t) \Delta t + \xi_1 \sqrt{2k_B T \Delta t} , \quad (1.28)$$

$$q_2(t + \Delta t) - q_2(t) = -\frac{\mu_{22}}{\kappa} q_2(t) \Delta t - \mu_{12} \phi'(q_1(t)) \Delta t + \xi_2 \sqrt{2k_B T \Delta t} \quad (1.29)$$

where ξ_i is a zero-mean Gaussian random variable, as

$$\langle \xi_i \xi_j \rangle = \mu_{ij} . \quad (1.30)$$

We choose to work in terms of mobility instead of friction matrix elements for convenience in the upcoming developments. They are, however, strictly equivalent.

Furthermore, in the rest of this study, we choose $\phi(q_1)$ to be sinusoidal, as $\phi(q_1) = \Delta U \sin\left(\frac{2\pi q_1}{\lambda}\right)$, with ΔU the amplitude of the sinusoidal trap.

1.4.2 Time step selection

In any discretisation process, it is essential to choose the appropriate time step. This time step must be small enough to avoid a blowup from error propagation (again, see the beginning of this thesis for more details on how the Euler method propagates errors), but big enough to remain advantageous. An arbitrarily small timestep may prohibit the probing of very long time dynamics due to computation time.

To select a correct time step for a simulation, I typically first identify all the smaller time scales in the system I want to simulate. In the present study, I identify two time

scales : the local relaxation time within a local minimum of the sinusoidal trap, and the relaxation time of the harmonic trap.

Near a local minimum, we may approximate the sinusoidal trap as harmonic. In that case,

$$\phi(q_1) = \frac{1}{2}k_{\text{eff}}q_1^2, \quad (1.31)$$

where k_{eff} is an effective stiffness, which we identify as the local curvature of the potential as

$$k_{\text{eff}} = |\phi''(0)| \quad (1.32)$$

$$= \Delta U \left(\frac{2\pi}{\lambda} \right)^2, \quad (1.33)$$

and we can finally identify this first timescale as

$$\tau_1 = (\mu_{11}k_{\text{eff}})^{-1} = \frac{\lambda^2}{4\pi^2\Delta U\mu_{11}}. \quad (1.34)$$

The harmonic trap is much simpler, as it requires no approximations, and we find its characteristic timescale

$$\tau_2 = \frac{\kappa}{\mu_{22}}. \quad (1.35)$$

Then, I choose a time step which satisfies

$$\Delta t = \frac{1}{100} \max \tau_1, \tau_2. \quad (1.36)$$

1.5 Results

1.6 Conclusion and perspectives on the chapter